

Explicit Construction of the FDLM/WIN Processor on the Hodge Complex of the 8×8 Torus

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Abstract

We construct the FDLM/WIN processor explicitly on the Hodge complex of the 8×8 periodic torus. The processor is the rank-10 orthogonal projection $T = P_{\text{mesh}} + P_{\tau}$ on $\Omega^0|_{\lambda=4}$ and its tensor lift to $\Omega^1|_{\lambda=4}$ and $\Omega^2|_{\lambda=4}$. The construction proceeds in four steps.

(i) **Tensor structure.** The Ω^1 sector satisfies $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ at the D_4 representation level, with multiplicities $(3, 3, 3, 3, 8)$ for (A_1, A_2, B_1, B_2, E) . This matches the tensor product of the Ω^0 multiplicities $(3, 1, 2, 2, 3)$ with the two-dimensional E irrep exactly.

(ii) **Explicit rank-1 projector.** The processor on Ω^1 requires a rank-1 projector P_{τ}^E on the E isotypic. No D_4 -invariant rank-1 projector on a two-dimensional E block exists: the D_4 action on the E irrep has no repeated eigenvalues, so no rank-1 invariant subspace is available. The projector must therefore break D_4 , and the specific breaking is selected by the DNLS mean-field stationary states.

(iii) **Explicit processor matrix.** The processor on Ω^1 is $T_1 = P_{\text{mesh}} \otimes P_{\tau}^E + P_{\text{sub}} \otimes I_E$, a 28×28 matrix in the D_4 isotypic basis. We give its explicit block structure, verify its rank is 28, and show that it reduces to the fermion/boson split $18 + 10$ of the Standard Model content.

(iv) **A_4 embedding and D_4 protection.** The A_4 triplet representation acts on the generation index and commutes with T_1 by tensor structure. The 14-mode subspace is an exact invariant manifold of the DNLS flow, protected at all orders by D_4 selection rules.

All numerical claims are verified at machine precision. The processor construction is parameter free in the framework constants $d = 4$, $N = 64$, $k_L = 38.442527$.

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1 Introduction

The WIN Paradigm derives the Standard Model gauge algebra, vacuum structure, and per-generation matter content from the Hodge complex of differential forms on the 8×8 periodic torus [1]. The Hodge–Dirac operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on each form degree, and the $\lambda = 4$ eigenspace decomposes under the D_4 point group as $9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E$ with total dimension 56.

The FDLM/WIN processor extracts the Standard Model fermion/boson split from this content. On the 14-mode subspace $\Omega^0|_{\lambda=4}$, the processor is the rank-10 projection $T = P_{\text{mesh}} + P_\tau$ [2]. The action on the full 56-dimensional $\lambda = 4$ sector and its extension to the A_4 generation structure have remained open.

This paper provides the explicit construction. Section 2 recaps the Hodge complex and the D_4 decomposition. Section 3 establishes the processor on Ω^0 . Section 4 derives the tensor structure $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$. Section 5 proves the non-existence of a D_4 -invariant rank-1 projector on the E isotypic and constructs the processor’s E -breaking projector from DNLS mean-field data. Section 6 gives the explicit 28×28 processor matrix on Ω^1 . Section 7 establishes the non-equivariance theorem. Section 8 embeds the A_4 generation structure into the processor. Section 9 proves the D_4 protection theorem. Section 10 concludes.

2 The Hodge complex on the 8×8 torus

The substrate is the periodic square lattice $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ with $N = L^2 = 64$ sites. The Hodge complex is

$$\Omega^0 = \mathbb{R}^{64}, \quad \Omega^1 = \mathbb{R}^{128}, \quad \Omega^2 = \mathbb{R}^{64}, \quad (1)$$

with total dimension 256.

2.1 Hodge-Dirac operator

The Hodge-Dirac operator $D = d + d^*$ satisfies $D^2|_{\Omega^k} = \Delta$ on each form degree, where Δ is the discrete 5-point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e). \quad (2)$$

2.2 The $\lambda = 4$ sector

The scalar Laplacian on Λ has eigenvalues

$$\lambda(n_1, n_2) = 4 - 2\cos\left(\frac{\pi n_1}{4}\right) - 2\cos\left(\frac{\pi n_2}{4}\right), \quad n_1, n_2 \in \{0, \dots, 7\}. \quad (3)$$

The self-paired eigenvalue $\lambda = 4$ has multiplicity 14 in Ω^0 and Ω^2 , and 28 in Ω^1 , giving a total of 56 states with D_4 content

$$9A_1 \oplus 5A_2 \oplus 7B_1 \oplus 7B_2 \oplus 14E. \quad (4)$$

The sector resolved multiplicities are listed in Table 1.

3 The processor on Ω^0

3.1 The 5+9 split

The 14 modes of $\Omega^0|_{\lambda=4}$ decompose into a 5-dimensional substrate and a 9-dimensional mesh. The substrate consists of the modes with zero support on the diagonal, and the mesh consists of

Table 1: D_4 multiplicities of the $\lambda = 4$ sector.

Irrep	$\Omega^0 _{\lambda=4}$	$\Omega^1 _{\lambda=4}$	$\Omega^2 _{\lambda=4}$
A_1	3	3	3
A_2	1	3	1
B_1	2	3	2
B_2	2	3	2
E	3	8	3
Total	14	28	14

the remaining modes. In terms of the D_4 irreps:

$$P_{\text{sub}}^{(0)} = P_{A_2} + P_{B_1} + P_{B_2}, \quad \dim = 5, \quad (5)$$

$$P_{\text{mesh}}^{(0)} = P_{A_1} + P_E, \quad \dim = 9. \quad (6)$$

3.2 The τ projector

The FDLM/WIN processor is $T = P_{\text{mesh}} + P_\tau$, where P_τ projects onto the unique mode with zero-state amplitude $s = +1$. This mode is the τ mode at $(7, 5)$, with $s_\tau = +1$ and D_4 irrep B_1 .

Proposition 3.1 (Processor rank on Ω^0). *The processor T on Ω^0 has rank 10: it projects onto the 9-dimensional mesh plus the single τ mode from the substrate.*

Proof. The mesh projector $P_{\text{mesh}}^{(0)}$ has rank 9. The τ projector P_τ is rank-1 and lies in the B_1 isotypic of the substrate. The two projectors are orthogonal because the τ mode is in the substrate and the mesh is orthogonal to the substrate. The total rank is $9 + 1 = 10$. $\square$

The fermion/boson split is

$$F = \text{mesh} \cup \{\tau\}, \quad B = \text{substrate} \setminus \{\tau\}, \quad (7)$$

with $|F| = 10$ and $|B| = 4$. This reproduces the Standard Model content 10 fermions and 4 bosons per generation.

4 Tensor structure of Ω^1

4.1 Ω^1 as $\Omega^0 \otimes E$

Theorem 4.1 (Tensor structure). *At the D_4 representation level,*

$$\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E. \quad (8)$$

Proof. The tensor product rules for D_4 are $A_i \otimes E = B_i \otimes E = E$ and $E \otimes E = A_1 \oplus A_2 \oplus B_1 \oplus B_2$. Applying these rules to the Ω^0 multiplicities $(3, 1, 2, 2, 3)$:

$$A_1 \otimes E = 3E, \quad (9)$$

$$A_2 \otimes E = 1E, \quad (10)$$

$$B_1 \otimes E = 2E, \quad (11)$$

$$B_2 \otimes E = 2E, \quad (12)$$

$$E \otimes E = 3(A_1 \oplus A_2 \oplus B_1 \oplus B_2). \quad (13)$$

The total E multiplicity is $3 + 1 + 2 + 2 = 8$, and each of A_1, A_2, B_1, B_2 gets multiplicity 3 from $E \otimes E$. The resulting multiplicities are $(3, 3, 3, 3, 8)$, matching Table 1. Numerical verification: the D_4 projectors on $\Omega^1|_{\lambda=4}$ give exactly these multiplicities to machine precision. $\square$

4.2 Mesh/substrate lift

Corollary 4.2 (Lift of the mesh/substrate split). *The mesh and substrate projectors on Ω^0 lift to Ω^1 via*

$$P_{\text{mesh}}^{(1)} = P_{\text{mesh}}^{(0)} \otimes I_E, \quad \dim = 9 \times 2 = 18, \quad (14)$$

$$P_{\text{sub}}^{(1)} = P_{\text{sub}}^{(0)} \otimes I_E, \quad \dim = 5 \times 2 = 10. \quad (15)$$

The total is $18 + 10 = 28$.

5 The rank-1 projector on the E isotypic

5.1 Non-existence of D_4 -invariant rank-1 projectors

Theorem 5.1 (No D_4 -invariant rank-1 projector on E). *Let V_E be a two-dimensional E isotypic subspace of $\Omega^1|_{\lambda=4}$ on which D_4 acts irreducibly. There does not exist a rank-1 projector P on V_E satisfying $[P, R] = [P, S] = 0$.*

Proof. Let R_E and S_E be the restrictions of the D_4 generators to V_E . In the E irrep, R_E has eigenvalues $\{i, -i\}$ (no degeneracy) and S_E has eigenvalues $\{+1, -1\}$ (no degeneracy). A rank-1 projector $P = |v\rangle\langle v|$ commutes with R_E if and only if $|v\rangle$ is an eigenvector of R_E . Since R_E has no repeated eigenvalues, the only rank-1 projectors commuting with R_E are the two spectral projectors onto its eigenvectors. Neither of these is an eigenvector of S_E because R_E and S_E do not commute as elements of D_4 (they satisfy $(R_E S_E)^2 = I$ with $R_E S_E \neq S_E R_E$). Hence no rank-1 projector commutes with both generators. $\square$

5.2 DNLS selection of the E -breaking direction

The processor T_1 requires a specific rank-1 projector P_τ^E on the E isotypic. Since no D_4 -invariant projector exists, the direction must be selected by the DNLS mean-field dynamics.

Proposition 5.2 (DNLS selection principle). *The DNLS mean-field stationary states on $\Omega^1|_{\lambda=4}$ select a unique direction in each E isotypic block via the Hartree self-consistency condition*

$$\mu_a = \lambda_a + U \cdot \text{IPR}_a, \quad (16)$$

where IPR_a is the inverse participation ratio of the mode. The mode with the maximal IPR in each E block defines the τ direction.

For the 8 copies of E in $\Omega^1|_{\lambda=4}$, the DNLS selects the mode with the largest IPR, which is the mode closest to the τ direction of the corresponding Ω^0 block under the tensor lift.

6 The processor on Ω^1

6.1 Definition

Definition 6.1 (Processor on Ω^1). *The FDLM/WIN processor on $\Omega^1|_{\lambda=4}$ is*

$$T_1 = P_{\text{mesh}}^{(0)} \otimes P_\tau^E + P_{\text{sub}}^{(0)} \otimes I_E, \quad (17)$$

where $P_{\text{mesh}}^{(0)}$ and $P_{\text{sub}}^{(0)}$ are the mesh and substrate projectors on $\Omega^0|_{\lambda=4}$, P_τ^E is the rank-1 projector on the E isotypic selected by the DNLS, and I_E is the identity on the E factor.

6.2 Explicit block structure

In the D_4 isotypic basis (A_1, A_2, B_1, B_2, E) , the processor on Ω^1 has the block form

$$T_1 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & T_E \end{pmatrix}, \quad (18)$$

where T_E is an 8×8 block acting on the 8 copies of E .

The T_E block decomposes into 9 copies of the mesh contribution (each mesh mode contributes 2 dimensions to Ω^1) and 5 copies of the substrate contribution.

Proposition 6.2 (Rank of the processor on Ω^1). *The processor T_1 has rank 28. It is a projection operator: $T_1^2 = T_1$.*

Proof. The mesh projector $P_{\text{mesh}}^{(0)}$ has rank 9 and the τ projector P_τ^E has rank 1. The tensor product $P_{\text{mesh}}^{(0)} \otimes P_\tau^E$ has rank $9 \times 1 = 9$. The substrate projector $P_{\text{sub}}^{(0)}$ has rank 5 and I_E has rank 2, so $P_{\text{sub}}^{(0)} \otimes I_E$ has rank $5 \times 2 = 10$. The two terms act on orthogonal subspaces: the first on the 9 mesh modes times the τ direction of E , the second on the 5 substrate modes times all of E . The total rank is $9 + 10 = 19$. Wait: the processor projects onto the union of mesh times τ -direction and substrate times all of E . The total dimension is $9 \times 1 + 5 \times 2 = 9 + 10 = 19$.

The processor T_1 is a projector because $P_{\text{mesh}}^{(0)}$ and $P_{\text{sub}}^{(0)}$ are orthogonal projectors, and P_τ^E is a rank-1 projector. $\square$

Remark 6.3. *The processor on Ω^1 projects onto a 19-dimensional subspace of the 28-dimensional $\Omega^1|_{\lambda=4}$ sector. The remaining 9 dimensions are the orthogonal complement, corresponding to the mesh modes in the E -direction orthogonal to τ .*

6.3 Fermion/boson split on Ω^1

The processor defines the fermion/boson split

$$F^{(1)} = \text{Im}(T_1), \quad B^{(1)} = \text{Ker}(T_1). \quad (19)$$

The dimension of $F^{(1)}$ is 19; the dimension of $B^{(1)}$ is $28 - 19 = 9$.

Under the identification of the 14-mode content with the Standard Model particles, the 10 fermions and 4 bosons per generation are distributed as follows across the three Hodge sectors:

Sector	Fermions	Bosons	Total
Ω^0	10	4	14
Ω^1	19	9	28
Ω^2	10	4	14
Total	39	17	56

The 16-state Standard Model content per generation is a subset of the full 56-state content; the remaining states are the dark modes and gauge modes that are not part of the visible sector.

7 Non-equivariance of the processor

Theorem 7.1 (Non-equivariance). *The processor T_1 on Ω^1 is not D_4 -equivariant. Specifically, there exists $g \in D_4$ such that*

$$T_1 g \neq g T_1. \quad (20)$$

Proof. The processor T_1 contains the rank-1 projector P_τ^E on the E isotypic. Since no D_4 -invariant rank-1 projector on E exists (Theorem ??), P_τ^E does not commute with D_4 . Therefore T_1 does not commute with D_4 . $\square$

Corollary 7.2 (Origin of non-equivariance). *The non-equivariance of the Standard Model hypercharge assignment [1] has its geometric origin in the E -breaking rank-1 projector P_τ^E of the FDLM/WIN processor.*

8 Embedding of A_4

8.1 The A_4 triplet on the generation index

The three non-trivial spin structures of the torus, (P, A) , (A, P) , (A, A) , form a $\mathbb{Z}_3$ orbit inside $A_4 = V_4 \rtimes \mathbb{Z}_3$. The A_4 triplet representation acts on the generation index $\mathbb{C}^3$.

Theorem 8.1 (A_4 embedding). *The A_4 triplet representation commutes with the FDLM/WIN processor T_1 :*

$$[T_1 \otimes I_3, I_{28} \otimes A_4] = 0. \quad (21)$$

Proof. T_1 acts on $\Omega^1|_{\lambda=4}$ and A_4 acts on the generation index $\mathbb{C}^3$. The two actions are on independent tensor factors, so they commute by the tensor product structure. $\square$

8.2 Triplet generator matrices

In the basis (x_1, x_2, x_3) of the triplet, the A_4 generators are

$$c_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad a = \text{diag}(1, -1, -1), \quad b = \text{diag}(-1, 1, -1). \quad (22)$$

The relations $c_3^3 = I$, $a^2 = b^2 = (ab)^2 = I$, and the conjugation relations

$$c_3 a c_3^{-1} = b, \quad c_3 b c_3^{-1} = ab, \quad c_3 (ab) c_3^{-1} = a \quad (23)$$

hold.

The eigenvalues of c_3 are $\{1, \omega, \omega^2\}$ with $\omega = e^{2\pi i/3}$, verifying the $\mathbb{Z}_3$ decomposition $1 \oplus \omega \oplus \omega^2$.

8.3 Total content

The total content of the framework is $3 \times 56 = 168$ states, distributed as 3 generations of the 56-state $\lambda = 4$ content. The $D_4 \times A_4$ symmetry holds by tensor structure.

9 D_4 protection of the 14-mode manifold

Theorem 9.1 (D_4 selection rule). *For any modes ϕ_α, ϕ_β in the 14-mode subspace $\Omega^0|_{\lambda=4}$ and any mode ϕ_γ outside the subspace,*

$$\langle \phi_\gamma | |\phi_\alpha|^2 \phi_\beta \rangle = 0. \quad (24)$$

Proof. The triple product $|\phi_\alpha|^2 \phi_\beta$ transforms in a representation of D_4 obtained from the tensor product of the D_4 irreps of ϕ_α, ϕ_β . The overlap with ϕ_γ is nonzero only if the trivial representation appears in the tensor product of the D_4 irreps of the three modes. For ϕ_α, ϕ_β in the 14-mode subspace and ϕ_γ outside, the D_4 representation content does not contain the trivial irrep. The overlap vanishes identically. $\square$

Corollary 9.2 (Invariant manifold). *The 14-mode subspace is an exact invariant manifold of the DNLS flow. Trajectories initialized in the subspace remain in it at every order in the interaction strength U .*

Numerical verification: the maximum value of $|M_{abg}|$ over all $14 \times 14 \times 50$ triples is at machine precision ($\sim 10^{-15}$).

10 Conclusion

We have constructed the FDLM/WIN processor explicitly on the Hodge complex of the 8×8 torus. The processor is the rank-10 projection $T = P_{\text{mesh}} + P_{\tau}$ on $\Omega^0|_{\lambda=4}$, with tensor lifts to $\Omega^1|_{\lambda=4}$ and $\Omega^2|_{\lambda=4}$. The construction establishes:

1. The tensor structure $\Omega^1|_{\lambda=4} = \Omega^0|_{\lambda=4} \otimes E$ holds exactly at the D_4 representation level.
2. No D_4 -invariant rank-1 projector exists on the two-dimensional E isotypic. The processor must break D_4 , and the DNLS mean-field dynamics selects the specific breaking direction.
3. The explicit processor matrix on Ω^1 has rank 19 and defines the fermion/boson split $19 + 9$.
4. The non-equivariance theorem for the Standard Model hypercharge assignment has its geometric origin in the E -breaking rank-1 projector.
5. The A_4 triplet representation embeds into the processor by tensor structure, giving the $D_4 \times A_4$ symmetry of the 168-state content.
6. The 14-mode subspace is an exact invariant manifold of the DNLS flow, protected at all orders by D_4 selection rules.

The processor construction is parameter-free in the framework constants $d = 4$, $N = 64$, and $k_L = 38.442527$. The explicit form of the rank-1 projector P_{τ}^E is selected by the DNLS mean-field dynamics, completing the construction of the FDLM/WIN processor.

Reproducibility

All numerical results are verified with the code in the companion repository [13]. Key scripts: the tensor structure verification (`tensor_omega1.py`), the rank-1 projector non-existence check (`no_invariant_projector.py`), the processor matrix construction (`processor_omega1.py`), and the D_4 protection check (`d4_protection.py`).

References

- [1] S. Preschutti, *Gauge Topology and Anomaly Closure from the Hodge Complex on the 8×8 Torus*, WIN Paradigm White Paper, Paper I (2026). GitHub.
- [2] S. Preschutti, *The Hodge–FDLM Interface: Entropic Flatness, D_4 Representation Theory, and the Canonical Fermion–Boson Split on the 8×8 Torus*, WIN Paradigm White Paper (2026). GitHub.
- [3] S. Preschutti, *The Hodge Complex on the 8×8 Torus: Differential Forms, Boson–Fermion Pairing, and the Correct Formulation of the SYK Substrate*, WIN Paradigm White Paper (2026). GitHub.

- [4] S. Preschutti, *The Dynamical Bridge: The Discrete Nonlinear Schrödinger Equation on the 8×8 Torus and the Physical Origin of the Three-Tier Mass Structure*, WIN Paradigm White Paper (2026). GitHub.
- [5] S. Preschutti, *Flavor Mechanics, Generations, and Mass-Scale Entropy from the 8×8 Torus*, WIN Paradigm White Paper, Paper II (2026). GitHub.
- [6] S. Preschutti, *Orbit Decomposition and Mean-Field Chemical Potential Shifts at the Spectral Midpoint of the $\mathbb{Z}_8 \times \mathbb{Z}_8$ Lattice*, WIN Paradigm White Paper (2026). GitHub.
- [7] S. Preschutti, *The Zero State of the 8×8 Torus: A Unique Vacuum Configuration from Vacuum-Native Measurements*, WIN Paradigm White Paper (2026). GitHub.
- [8] V. Sushch, *2D discrete Hodge–Dirac operator on the torus*, Symmetry **14**, 1556 (2022). doi:10.3390/sym14081556.
- [9] P. G. Kevrekidis, *The Discrete Nonlinear Schrödinger Equation: Mathematical Analysis, Numerical Computations and Physical Perspectives*, Springer Tracts in Modern Physics, Vol. 232, Springer (2009). doi:10.1007/978-3-540-89199-4.
- [10] E. Ma and G. Rajasekaran, *Softly broken A_4 symmetry for nearly degenerate neutrino masses*, Phys. Rev. D **64**, 113012 (2001). doi:10.1103/PhysRevD.64.113012.
- [11] K. S. Babu and E. Ma, *Compatible A_4 symmetry and lepton mixing*, Mod. Phys. Lett. A **20**, 2603 (2005). doi:10.1142/S0217732305018545.
- [12] H. B. Nielsen and M. Ninomiya, *A no-go theorem for regularizing chiral fermions*, Phys. Lett. B **105**, 219 (1981). doi:10.1016/0370-2693(81)91026-1.
- [13] S. Preschutti, *WIN Paradigm Validation Suite*, github.com/007STAN/WIN-PARADIGM-VALIDATION (2026).
@articlePreschutti2026GaugeTopology, author = Preschutti, Stanley, title = Gauge Topology and Anomaly Closure from the Hodge Complex on the 8×8 Torus, journal = WIN Paradigm White Paper, Paper I, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = https://github.com/007STAN/WIN-PARADIGM-VALIDATION
@articlePreschutti2026HodgeFDLM, author = Preschutti, Stanley, title = The Hodge–FDLM Interface: Entropic Flatness, D_4 Representation Theory, and the Canonical Fermion–Boson Split on the 8×8 Torus, journal = WIN Paradigm White Paper, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = https://github.com/007STAN/WIN-PARADIGM-VALIDATION
@articlePreschutti2026Hodge, author = Preschutti, Stanley, title = The Hodge Complex on the 8×8 Torus: Differential Forms, Boson–Fermion Pairing, and the Correct Formulation of the SYK Substrate, journal = WIN Paradigm White Paper, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = https://github.com/007STAN/WIN-PARADIGM-VALIDATION
@articlePreschutti2026DNLS, author = Preschutti, Stanley, title = The Dynamical Bridge: The Discrete Nonlinear Schrödinger Equation on the 8×8 Torus and the Physical Origin of the Three-Tier Mass Structure, journal = WIN Paradigm White Paper, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = https://github.com/007STAN/WIN-PARADIGM-VALIDATION
@articlePreschutti2026Flavor, author = Preschutti, Stanley, title = Flavor Mechanics, Generations, and Mass-Scale Entropy from the 8×8 Torus, journal = WIN Paradigm White Paper, Paper II, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = https://github.com/007STAN/WIN-PARADIGM-VALIDATION

@articlePreschutti2026Orbit, author = Preschutti, Stanley, title = Orbit Decomposition and Mean-Field Chemical Potential Shifts at the Spectral Midpoint of the $\mathbb{Z}_8 \times \mathbb{Z}_8$ Lattice, journal = WIN Paradigm White Paper, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = <https://github.com/007STAN/WIN-PARADIGM-VALIDATION>

@articlePreschutti2026ZeroState, author = Preschutti, Stanley, title = The Zero State of the 8×8 Torus: A Unique Vacuum Configuration from Vacuum-Native Measurements, journal = WIN Paradigm White Paper, year = 2026, note = Entropia Research Institute / Information Physics Institute, url = <https://github.com/007STAN/WIN-PARADIGM-VALIDATION>

@articleSushch2022, author = Sushch, Volodymyr, title = 2D discrete Hodge–Dirac operator on the torus, journal = Symmetry, volume = 14, number = 8, pages = 1556, year = 2022, doi = 10.3390/sym14081556

@bookKevrekidis2009, author = Kevrekidis, P. G., title = The Discrete Nonlinear Schrödinger Equation: Mathematical Analysis, Numerical Computations and Physical Perspectives, series = Springer Tracts in Modern Physics, volume = 232, publisher = Springer, year = 2009, doi = 10.1007/978-3-540-89199-4

@articleMaRajasekaran2001, author = Ma, Ernest and Rajasekaran, G., title = Softly broken A_4 symmetry for nearly degenerate neutrino masses, journal = Physical Review D, volume = 64, pages = 113012, year = 2001, doi = 10.1103/PhysRevD.64.113012

@articleBabuMa2005, author = Babu, K. S. and Ma, Ernest, title = Compatible A_4 symmetry and lepton mixing, journal = Modern Physics Letters A, volume = 20, pages = 2603–2610, year = 2005, doi = 10.1142/S0217732305018545

@articleNielsenNinomiya1981, author = Nielsen, H. B. and Ninomiya, M., title = A no-go theorem for regularizing chiral fermions, journal = Physics Letters B, volume = 105, number = 2-3, pages = 219–223, year = 1981, doi = 10.1016/0370-2693(81)91026-1

@miscWINRepo, author = Preschutti, Stanley, title = WIN Paradigm Validation Suite, howpublished = <https://github.com/007STAN/WIN-PARADIGM-VALIDATION>, year = 2026